

Notes: Farber and Gibbons (QJE, 1996)

Learning and Wage Dynamics

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1 Big picture

- **Question.** How do wages evolve when employers gradually learn about worker ability from observed performance? The paper asks whether a competitive labor market with public learning can explain key facts about early-career wage dynamics.
- **Core idea.** At labor-market entry, employers observe schooling and some worker characteristics, but they do not perfectly know productive ability. As performance signals accumulate, the market's posterior belief about ability changes, and wages adjust because wages equal expected output.
- **Benchmark model.** The authors build a public-learning model: all employers observe the same worker characteristics and the same subsequent output signals. This contrasts with private-learning models in which only the current employer observes performance.
- **Three main predictions.**
 1. The estimated effect of schooling on wage levels should be independent of labor-market experience, even though schooling becomes less important in the market's inference process over time.
 2. Time-invariant variables correlated with ability but not observed by employers should become increasingly correlated with wages as experience accumulates.
 3. Wage residuals should follow a martingale, or approximately a martingale once measurement error is added.
- **Empirical setting.** The authors use the National Longitudinal Survey of Youth (NLSY), focusing on young workers making their first long-term transition into the labor market.
- **Main empirical result.** The evidence is broadly consistent with employer learning: schooling does not have a significant interaction with experience in wage-level regressions, while the AFQT residual and library-card residual become more strongly related to wages with experience.
- **Qualification.** The strict martingale covariance restrictions are statistically rejected, but the qualitative pattern is close to the theory. A richer AR(1) learning model fits the covariance structure substantially better.
- **Economic takeaway.** Wage growth is not only human-capital accumulation. Part of early-career wage dynamics reflects the market learning about worker ability and incorporating new signals into wages.

2 Incremental contribution of the paper

- **From signaling at entry to dynamic learning after entry.** Earlier signaling models emphasize how schooling may communicate information at labor-market entry. This paper studies how that initial signal is gradually supplemented by observed performance after entry.
- **A tractable benchmark for employer learning.** The paper provides a simple public-learning benchmark that is analytically transparent, works for many distributions and production functions, and yields testable restrictions.

- **A key separation between inference and regression coefficients.** One of the paper’s most important insights is that schooling can become less important in employers’ posterior inference while the estimated level-wage coefficient on schooling remains constant with experience.
- **Use of market-unobserved ability proxies.** The paper uses residualized AFQT scores and family-background information as variables correlated with ability but not directly observed by the market. This creates a direct empirical test of gradual learning.
- **Covariance-structure evidence.** Beyond wage-level regressions, the paper tests whether residuals obey the martingale restrictions implied by learning. This connects employer learning to the full time-series structure of wage residuals.
- **Bridge between human-capital and information models.** The paper does not reject human capital. Instead, it argues that learning and human capital are complementary mechanisms and that future wage-dynamics models should combine learning, mobility, and training.
- **Methodological contribution.** By working in wage levels rather than log wages, the paper aligns the empirical specification with the model’s conditional-expectation structure. This is important because the main predictions are about wage levels.

3 Data and measurement

3.1 NLSY sample

- The empirical analysis uses the National Longitudinal Survey of Youth (NLSY), which follows individuals who were young at the beginning of the survey.
- The NLSY is especially useful because employer learning should be most visible early in the career, when ability is still uncertain and performance signals are arriving.
- The authors focus on individuals who make a first long-term transition from nonwork to work during the survey period.
- A worker is classified as having made a long-term transition if the worker spends three consecutive years primarily working after at least one year not primarily working.
- The final construction produces 4,998 individuals for the broader covariance analysis and 4,970 individuals with 28,984 wage observations for the main earnings regressions.

Interpretation: The sample is designed to capture the start of the learning process. Rather than using potential experience, the authors construct an experience measure based on actual transition into stable work.

3.2 Wages and experience

- The wage measure is real hourly earnings on the current job, deflated to 1982–1984 dollars.
- Experience is measured as time since the worker’s first full-time entry into the labor force according to the authors’ transition rule.
- Table I shows that real wages rise strongly with experience: the mean wage rises from about 4.96 at experience 0 to roughly 7.88 by experience 11.

- Average education is roughly stable around 13 years across experience cells, which helps separate experience effects from education composition effects.

3.3 Ability proxies not observed by the market

- The empirical strategy needs variables correlated with ability but not directly observed by employers.
- The first proxy is the AFQT score, constructed from components of the Armed Forces Vocational Aptitude Battery.
- The second proxy is whether the household had a library card when the respondent was age fourteen.
- The paper residualizes these variables against observables and the first observed wage, producing an AFQT residual and a library-card residual.
- The purpose of residualizing on the first wage is to purge information that the market had already incorporated into the initial wage.

Interpretation: These residual variables are designed to capture ability-related information that the econometrician observes but the labor market has not yet fully learned.

4 Models and empirical specifications

4.1 Worker output and public learning

Let η_i denote worker ability, s_i schooling, X_i characteristics observed by employers and the econometrician, Z_i characteristics observed by employers but not in the data, and B_i background variables in the data but not directly observed by employers. Output in period t is drawn from a conditional distribution,

$$y_{it} \sim G(y_{it} \mid \eta_i, s_i, X_i, Z_i).$$

- Employers observe schooling and other public characteristics at entry.
- Employers then observe performance signals over time.
- All employers observe the same information, so learning is public rather than private.

4.2 Wage determination

The central wage-setting assumption is

$$w_{it} = E[y_{it} \mid s_i, X_i, Z_i, y_{i1}, \dots, y_{i,t-1}].$$

- Wages equal expected output given all public information available at date t .
- There are no long-term contracts in the benchmark model.
- Wage changes reflect updates in the market's posterior belief about ability.

4.3 Prediction 1: schooling coefficient does not vary with experience

The paper considers wage-level regressions of the form

$$w_{it} = \alpha_t + \beta_t s_i + X_i \gamma_t + \varepsilon_{it}.$$

Because wages are conditional expectations and output draws are identically distributed conditional on ability and observables, the law of iterated projections implies

$$E^*[w_{it} \mid s_i, X_i] = E^*[y_{it} \mid s_i, X_i],$$

where the right-hand side is independent of t . Therefore the estimated coefficient on schooling should not change with experience.

Economic intuition: Schooling becomes less useful for inference as actual performance is observed, but the surprise component of later wages is orthogonal to the information used to set initial wages. Hence the level-wage coefficient on schooling remains constant.

4.4 Prediction 2: market-unobserved ability proxies gain predictive power

Let B_i^* be the residual from projecting an ability proxy on observables and the first wage:

$$B_i^* = B_i - E^*[B_i \mid s_i, X_i, w_{i0}].$$

The empirical regression adds B_i^* and its interaction with experience:

$$w_{it} = \alpha + \beta s_i + X_i \gamma + \pi B_i^* + \rho(B_i^* \times t) + \varepsilon_{it}.$$

The learning model predicts $\rho > 0$ if B_i^* is positively correlated with ability.

Economic intuition: The market does not initially observe B_i^* . But as output signals arrive, wages increasingly reflect ability. Since B_i^* is correlated with ability, its wage correlation rises with experience.

4.5 Prediction 3: wage residuals are a martingale

After controlling for the systematic experience profile and observables, the learning model implies that residual changes should not be predictable using prior residuals:

$$E[\varepsilon_{it} \mid \varepsilon_{i,t-1}, \varepsilon_{i,t-2}, \dots] = \varepsilon_{i,t-1}.$$

With measurement error, observed wage residuals can be written as a martingale component plus transitory noise. This imposes strong restrictions on the within-worker residual covariance matrix.

4.6 Earnings-function regression

The main level-wage regression estimated in Table II is conceptually

$$\begin{aligned} w_{it} = & \alpha_0 + \alpha_1 \text{Experience}_{it} + \alpha_2 \text{Experience}_{it}^2 + \beta \text{Education}_i \\ & + \theta(\text{Education}_i \times \text{Experience}_{it}) + \pi_1 \text{AFQT}_i^* + \rho_1(\text{AFQT}_i^* \times \text{Experience}_{it}) \\ & + \pi_2 \text{Library}_i^* + \rho_2(\text{Library}_i^* \times \text{Experience}_{it}) + \text{Controls}_{it} + \varepsilon_{it}. \end{aligned}$$

- The model predicts $\theta = 0$.
- The model predicts $\rho_1 > 0$ and $\rho_2 > 0$.
- The specification includes calendar-year controls and education-by-year controls to absorb the changing return to education in the 1980s.

5 Identification and empirical design

5.1 What identifies learning effects?

- The key identification comes from how the relationship between wages and fixed worker characteristics changes with labor-market experience.
- If employers learn about ability through output signals, then market-unobserved ability proxies should matter more for wages later in the career.
- If education is simply an initial signal whose regression coefficient is preserved by iterated expectations, then the education-by-experience coefficient should be zero in wage-level regressions.

5.2 Why residualize AFQT and library-card variables?

- Raw AFQT scores and household background variables may be correlated with characteristics observed by employers.
- The paper therefore residualizes these variables using schooling, demographics, part-time status, calendar year, and the first observed wage.
- The first observed wage is intended to summarize the market's initial information about ability.

Identification concern: If the first wage is measured with error, the residualized ability proxies may still contain information that employers already knew. This could explain why the AFQT residual has a positive main effect even at low experience.

5.3 Human capital versus learning

- A pure human-capital model can explain rising wages with experience but does not naturally generate all three learning predictions simultaneously.
- If on-the-job training interacts with schooling, the education-by-experience coefficient should generally be nonzero.
- If on-the-job training interacts with ability, then ability proxies may become more important with experience even without learning.
- The covariance restrictions help distinguish learning from such alternatives, because the martingale structure is a distinctive implication of learning.

5.4 Limitations of the design

- The public-learning assumption is strong. In many real workplaces, current employers may observe more information than outside firms.
- The benchmark model abstracts from job mobility and rich human-capital investment.
- The wage-level specification follows the model but differs from the more common log-wage specification.
- The martingale prediction is statistically rejected, implying that the benchmark model is incomplete.

Interpretation: The paper is best read as evidence that public learning is an important component of wage dynamics, not as evidence that learning alone fully explains wages.

6 Tables and figures

6.1 Table I: Summary Statistics, NLSY, by Experience

Read:

- Table I reports means and standard deviations by years of labor-market experience.
- The sample includes 34,742 observations on 4,998 workers.
- Mean real wages rise strongly with experience, from 4.96 at experience 0 to around 7.88 at experience 11.
- Average education remains near 13 years across experience levels, which suggests that the sample composition by education is not changing mechanically in a way that explains the wage growth.
- Marriage rates rise with experience, from 0.198 at experience 0 to roughly 0.570 by experience 11.
- The table documents the young-worker setting where learning should be empirically most visible.

Economic message: The data show rapid early-career wage growth and changing household status, but relatively stable education by experience. This motivates the learning analysis and the use of careful controls in the wage regressions.

TABLE I
SUMMARY STATISTICS
NLSY
(MEANS AND STANDARD DEVIATIONS)
ALL YEARS OF EMPLOYMENT (1979–1991) BY EXPERIENCE

Experience	N	Wage	Age	Education	Part-Time	Coll. barg.	Nonwhite	Female	Married	Marr. & female
0	1169	4.96 (2.02)	22.2 (2.56)	12.5 (2.33)	0.0	.170	.441	.483	.198	.119
1	4589	5.33 (2.33)	22.7 (2.68)	12.9 (2.35)	.132	.165	.415	.506	.202	.124
2	4622	5.90 (2.62)	23.6 (2.67)	13.0 (2.37)	.087	.180	.418	.504	.270	.162
3	4623	6.30 (2.86)	24.6 (2.67)	13.0 (2.39)	.071	.186	.421	.504	.332	.188
4	4182	6.70 (3.13)	25.5 (2.51)	13.1 (2.37)	.090	.196	.413	.505	.387	.213
5	3749	7.01 (3.34)	26.2 (2.38)	13.1 (2.40)	.096	.190	.416	.497	.441	.233

6	3269	7.27 (3.48)	27.0 (2.25)	13.1 (2.36)	.094	.185	.407	.501	.479	.252
7	2740	7.53 (3.62)	27.6 (2.10)	13.1 (2.31)	.093	.191	.397	.490	.510	.262
8	2170	7.77 (3.61)	28.3 (1.97)	13.0 (2.29)	.102	.191	.393	.486	.541	.270
9	1640	7.89 (3.74)	29.0 (1.90)	13.0 (2.29)	.098	.189	.387	.493	.573	.285
10	1230	7.77 (3.64)	29.8 (1.79)	12.9 (2.21)	.104	.196	.412	.496	.566	.281
11	759	7.88 (3.68)	30.6 (1.66)	12.9 (2.17)	.083	.202	.406	.511	.570	.283
Total:	34,742	6.64 (3.21)	25.6 (3.27)	13.0 (2.35)	.092	.184	.412	.500	.386	.207

The numbers in parentheses are standard deviations. The Part-time, Collective bargaining, Nonwhite, Female, Married, and Married & female variables are dummy variables. Wage data are in real 1982–1984 dollars (deflated by CPI). Observations at the time of entry (experience = 0) which are part-time are not included in this analysis. See text for details.

(a) (a) Table I, part 1: experience 0–5

(b) (b) Table I, part 2: experience 6–11 and notes

6.2 Table II: Regression Analysis of Earnings Function

Read:

- Table II is the paper’s central wage-regression table.
- Column (2) estimates a level-wage regression with experience, experience squared, education, and education interacted with experience.
- Column (3) adds AFQT and library-card residuals and their interactions with experience.
- Column (4) adds demographic controls including age at entry, part-time status, collective bargaining, race, sex, marital status, and sex-by-marital-status.
- In columns (2)–(4), the education-by-experience coefficient is essentially zero and statistically insignificant.
- The AFQT residual-by-experience coefficient is positive and significant; the library-card residual-by-experience coefficient is also positive and significant.
- Column (5), which omits year and education-by-year controls, produces a positive education-by-experience coefficient, showing why controlling for rising returns to education in the 1980s is important.
- Column (6) uses log wages and shows a negative education-by-experience coefficient, consistent with the authors’ point that the theory is about wage levels, not log wages.

Economic message: Table II supports the two main cross-sectional learning predictions: schooling does not gain or lose predictive power in level-wage regressions as experience rises, while ability proxies not observed by the market become increasingly related to wages.

Independent variable	(1) Mean [sd]	(2) Wage (level)	(3) Wage (level)	(4) Wage (level)	(5) Wage (level)	(6) Wage (log)
Constant	1.0	-3.5579 (0.785)	-3.8086 (0.788)	-6.0321 (0.928)	-2.7034 (0.388)	0.0873 (0.124)
Experience	5.1804 [2.502]	0.4428 (0.102)	0.5054 (0.103)	0.5366 (0.100)	0.2897 (0.069)	0.1012 (0.013)
Experience squared	33.0953 [29.947]	-0.0178 (0.003)	-0.0185 (0.003)	-0.0178 (0.003)	-0.0198 (0.003)	-0.0027 (0.000)
Education	13.0450 (2.349)	0.6745 (0.061)	0.6938 (0.061)	0.6719 (0.059)	0.4602 (0.024)	0.0969 (0.007)
Education × experience	67.5424 (35.014)	-0.0004 (0.008)	-0.0049 (0.008)	-0.0041 (0.007)	0.0172 (0.005)	-0.0026 (0.001)

AFQT residual/100	0.0024 [0.148]	—	0.6494 (0.307)	0.8734 (0.291)	0.7841 (0.292)	0.1880 (0.044)
AFQT resid/100 × experience	0.0189 [0.856]	—	0.1938 (0.064)	0.1848 (0.060)	0.1922 (0.060)	0.0187 (0.008)
Lib card resid/10	-0.0002 [0.043]	—	0.2583 (1.035)	0.2130 (0.988)	-0.0579 (0.989)	0.1440 (0.146)
Lib card resid × experience/10	-0.00011 [0.248]	—	0.6035 (0.205)	0.6169 (0.192)	0.6448 (0.192)	0.0588 (0.026)
Year		yes	yes	yes	no	yes
Education × year		yes	yes	yes	no	yes
Other demographic		no	no	yes	yes	yes
R ²		0.215	0.224	0.294	0.289	0.296

The dependent variable is real hourly earnings on the current job (in levels in columns (2)–(5) and in logs in column (6)). The mean of the level of earnings is 8.91 (s.d. = 2.30). The mean of the log of earnings is 1.83 (s.d. = 0.448). The numbers in parentheses are Whitefisher standard errors computed accounting for the fact that there are multiple observations for each worker. There are 25,564 wage observations on 4970 individuals. Where indicated, there are ten year dummies for 1981–1990 and interactions of education with each of the ten year dummies. The base year is 1991. The other demographic characteristics, where included, consist of age at entry, a dummy variable for part-time, the interaction of part-time with education, and dummy variables for collective bargaining coverage, race, sex, marital status, and the interaction of sex and marital status.

(a) (a) Table II, part 1: baseline variables

(b) (b) Table II, part 2: residual interactions and fit

6.3 Table III: Empirical Covariance Matrix of Wage Residuals

Read:

- Table III reports the within-worker covariance matrix of wage residuals in levels.
- Residuals come from separate OLS regressions by year-experience cell, controlling for age, education, part-time status, union coverage, race, sex, marital status, and sex-by-marital-status.
- The diagonal elements rise with experience: for example, the variance is 2.58 at experience 0, 6.83 at experience 4, and 10.0 at experience 11.
- Off-diagonal covariances generally rise as two residuals share more common prior wage innovations.

- However, covariances within a column decline as one moves down the column, which violates the strict martingale-with-classical-measurement-error structure.

Economic message: Table III provides qualitative support for learning because residual variance and covariance grow with accumulated information. But the exact covariance restrictions are too strong, suggesting the benchmark model needs enrichment.

TABLE III
EMPIRICAL COVARIANCE MATRIX OF WITHIN WORKER WAGE RESIDUALS (LEVELS)
(STANDARD ERRORS)
(CELL SIZE)

	0	1	2	3	4	5	6	7	8	9	10	11
0	2.58 (.389) [1169]											
1	1.82 (.379) [1081]	3.85 (.395) [1589]										
2	1.72 (.391) [1080]	2.45 (.335) [1292]	4.81 (.419) [1622]									
3	1.96 (.397) [1092]	2.45 (.346) [1282]	3.40 (.372) [1433]	5.67 (.443) [1623]								
4	1.57 (.400) [966]	2.32 (.354) [1075]	3.27 (.376) [1399]	4.11 (.399) [1487]	6.83 (.487) [1853]							
5	1.45 (.400) [966]	2.08 (.354) [1075]	2.95 (.376) [1399]	3.82 (.399) [1487]	4.74 (.487) [1853]	7.12 (.500) [1900]						

6	1.67 (.444) [705]	2.17 (.373) [505]	3.03 (.411) [507]	3.75 (.426) [507]	4.73 (.481) [507]	5.16 (.477) [507]	8.42 (.562) [507]					
7	1.58 (.527) [567]	2.31 (.396) [529]	2.89 (.435) [568]	3.53 (.450) [568]	4.36 (.488) [568]	4.91 (.498) [568]	6.49 (.563) [568]	9.14 (.600) [568]				
8	1.40 (.515) [419]	1.84 (.405) [3006]	2.67 (.449) [3047]	3.11 (.451) [3051]	3.84 (.481) [3016]	4.35 (.504) [3016]	5.88 (.565) [3016]	6.59 (.580) [3016]	9.30 (.628) [3016]			
9	1.04 (.522) [316]	1.73 (.434) [1513]	2.54 (.455) [1543]	2.75 (.464) [1548]	3.70 (.510) [1521]	3.94 (.532) [1530]	5.38 (.586) [1519]	6.16 (.599) [1511]	7.32 (.634) [1495]	9.97 (.689) [1640]		
10	0.825 (.518) [240]	1.35 (.403) [1125]	2.33 (.495) [1154]	2.52 (.490) [1156]	3.22 (.549) [1125]	3.60 (.543) [1133]	4.64 (.599) [1151]	5.29 (.630) [1132]	5.89 (.632) [1105]	6.92 (.640) [1112]	8.97 (.714) [1230]	
11	0.566 (.407) [148]	1.24 (.486) [983]	2.22 (.579) [704]	2.17 (.558) [711]	3.12 (.619) [696]	2.93 (.592) [695]	4.41 (.664) [706]	4.68 (.686) [718]	5.53 (.703) [678]	6.33 (.757) [669]	6.72 (.766) [704]	10.0 (.853) [759]

Based on residuals from separate OLS regressions for each year-experience cell of the real wage on dummy variables for age (all values), education category (four values), part-time status, the interaction of the education dummies with part-time status, collective bargaining status, nonwhite, female, marital status, and the interaction of female and marital status. The elements of the covariance matrix are computed as defined in equation (A1), and the standard errors of the elements of the covariance matrix are computed as defined in equation (A2).

- (a) (a) Table III, part 1: upper-left covariance block (b) (b) Table III, part 2: lower rows and notes

6.4 Table IV: OMD Estimation of Martingale Covariance Structure

Read:

- Table IV estimates the covariance structure implied by a martingale residual process overlaid with classical measurement error.
- The estimated variance of initial unmeasured expected ability is 2.0404.
- The estimated variance of measurement error is 1.5704.
- The variance of wage innovations is positive in every period, consistent with the idea that new information continues to arrive over time.
- The structural test statistic is 157 with 54 degrees of freedom and a p-value near 1×10^{-11} , so the exact martingale covariance restrictions are rejected.
- Despite rejection, the model accounts for a large share of the qualitative variation in the covariance matrix, according to the discussion in the text.

Economic message: The martingale model captures the direction of the covariance patterns but not the exact shape. The rejection is important because it motivates richer learning models, especially the AR(1) ability process.

TABLE IV
OPTIMAL MINIMUM DISTANCE ESTIMATION OF COVARIANCE STRUCTURE MARTINGALE
OVERLAID WITH CLASSICAL MEASUREMENT ERROR NLSY UNBALANCED PANELS
(STANDARD ERRORS IN PARENTHESES)

Name	Parameter	Estimate
Variance of initial unmeasured expected ability	σ_1^2	2.0404 (0.089)
Variance of measurement error	σ_ϕ^2	1.5704 (0.058)
Variance of wage innovations each period:	$\sigma_{\mu 1}^2$	0.8634 (0.080)
	$\sigma_{\mu 2}^2$	0.5986 (0.082)
	$\sigma_{\mu 3}^2$	0.8748 (0.096)
	$\sigma_{\mu 4}^2$	0.5522 (0.094)
	$\sigma_{\mu 5}^2$	1.3158 (0.141)
	$\sigma_{\mu 6}^2$	0.8263 (0.146)
	$\sigma_{\mu 7}^2$	0.7643 (0.161)
	$\sigma_{\mu 8}^2$	0.6568 (0.200)
	$\sigma_{\mu 9}^2$	0.2891 (0.265)
	$\sigma_{\mu 10}^2$	0.8894 (0.476)
χ^2 statistic, structural test:		157
Degrees of freedom		54
p-value of test statistic		1×10^{-11}
Number of workers		4998

The optimal minimum distance estimator is the GLS regression of the unique elements of the covariance matrix in Table III on the variables implied by (A3). The covariance matrix of the covariance elements in Table III is defined in (A4). The χ^2 test statistic is the minimized value of the quadratic form in (A5), and the degrees of freedom are the number of restrictions (A3) places on the unrestricted covariance matrix in Table III.

(a) Table IV: OMD martingale covariance model

6.5 Table V: Average Age and Education by Entry Cohort and Sex

Read:

- Table V describes the final sample by entry cohort and sex.
- The full sample has 4,998 workers with average entry age 21.7 and average education 12.9 years.
- Earlier entry cohorts are younger and less educated, while later cohorts enter at older ages and with more schooling.
- Men and women are similar in age and education, though women are slightly older and slightly more educated on average at entry.
- The table documents why the authors use careful controls for cohort and education composition.

Economic message: The sample is not a simple cross-section of identical labor-market entrants. Entry cohort, age, and education vary systematically, so the paper's identification relies on within-worker panel structure and flexible controls.

TABLE V
AVERAGE AGE AND EDUCATION BY ENTRY COHORT AND SEX

Entry cohort	N	Average age	Average education
Overall			
1979	673	19.6	12.1
1980	576	19.8	12.5
1981	638	20.1	12.5
1982	550	20.9	12.8
1983	607	21.4	13.0
1984	588	22.4	13.0
1985	478	23.3	13.4
1986	407	24.1	13.7
1987	309	25.0	13.6
1988	172	26.2	13.5
All	4998	21.7	12.9
Males			
Entry cohort	N	Average age	Average education
1979	339	19.4	11.8
1980	261	19.6	12.1
1981	340	20.0	12.2
1982	282	20.7	12.4
1983	309	21.2	12.7
1984	296	22.1	12.7
1985	233	23.3	13.5
1986	206	24.3	13.6
1987	139	25.2	13.8
1988	79	26.3	13.7
All	2484	21.5	12.7
Females			
Entry cohort	N	Average age	Average education
1979	334	19.7	12.5
1980	315	20.1	12.8
1981	298	20.2	12.8
1982	268	21.1	13.2
1983	298	21.6	13.3
1984	292	22.8	13.4
1985	245	23.3	13.2
1986	201	24.3	13.7
1987	170	25.2	13.5
1988	93	26.3	13.3
All	2514	21.9	13.1

(a) Table V: age and education by entry cohort and sex

7 Short summary

- The paper develops a public-learning model in which wages equal expected output and employers learn worker ability from observed performance.
- The model predicts that schooling's level-wage coefficient does not vary with experience.
- The model predicts that ability-related variables not initially observed by employers become more correlated with wages over time.

- The model predicts that wage residuals should be a martingale, with predictable covariance restrictions once measurement error is added.
- Using NLSY data, the authors find no significant education-by-experience interaction in wage-level regressions when controlling for calendar-year changes in the return to schooling.
- AFQT and library-card residuals interact positively with experience, consistent with employers gradually learning ability.
- The martingale covariance structure is formally rejected, but the covariance patterns are qualitatively consistent with learning.
- A richer model in which worker ability follows an AR(1) process fits the residual covariance matrix much better.

8 Conclusion

8.1 Core conclusion

Farber and Gibbons show that employer learning provides a powerful and parsimonious explanation for important patterns in early-career wages. Wages behave as if the labor market gradually incorporates information about worker ability.

8.2 Economic intuition

At labor-market entry, schooling is a noisy signal of ability. As performance is observed, employers update beliefs. Variables that employers do not initially observe but that are correlated with ability should therefore become more predictive of wages with experience. This is exactly the pattern the paper documents with residualized AFQT and library-card measures.

8.3 Incremental contribution revisited

The paper’s key contribution is not just that ability matters for wages. Its deeper contribution is to show how the information structure of the labor market implies testable restrictions on wage-level regressions and residual covariance matrices.

8.4 Implications for labor economics

- Education can be both productive and informational; the paper focuses on its informational role in early-career wage dynamics.
- Test scores and family-background variables may matter more over time not because employers observe them, but because employers learn the ability they proxy.
- Wage growth reflects both human-capital accumulation and information revelation.
- Empirical wage equations should distinguish calendar-time changes in returns to education from genuine experience interactions.

8.5 Limitations

- The public-learning assumption abstracts from private employer information.
- The model largely abstracts from job mobility, promotions, and firm-specific matching.
- Human-capital acquisition is modeled simply, so the framework cannot fully explain the experience-earnings profile.
- The strict martingale covariance restrictions are rejected, indicating that a richer dynamic model is needed.

8.6 Takeaway

Learning is a central mechanism in wage dynamics. The labor market does not know worker ability at entry, and wages evolve as signals arrive. The paper's evidence from the NLSY shows that this learning channel is empirically visible and quantitatively important, even though it must ultimately be combined with models of human capital and mobility.